

Vibhu Singh

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EDUCATION

University of Maryland, College Park

College Park, MD

BS in Computer Science, Minor in Statistics; GPA: 3.7

Aug. 2022– May 2026

- **Relevant Coursework:** Introduction to Artificial Intelligence, Object Oriented Programming, Computer Systems, Discrete Systems, Calculus I and II, Linear Algebra, Functional Programming, Data Science, Algorithms, Applied Probability and Statistics I and II, Advanced Data Structures, Programming Handheld Systems, Sampling Theory, Foundations of Cybersecurity, Using Databases with Python, Introduction to Machine Learning

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, JavaScript, R, HTML, CSS, OCaml, Rust, SAS, MIPS, Rust, Dart

Libraries and Frameworks: Git, Flutter, Snowflake, Flask, React, Node.js, Linux, AWS, Redis, MongoDB, PostgreSQL

EXPERIENCE

DevOps Intern

June 2025 – Present

Chryselys

Palo Alto, CA

- Designed and developed scalable, configuration-driven data pipelines using Python and AWS, supporting ingestion, transformation, and publishing of data for internal product solutions.
- Collaborated cross-functionally with product and core development teams to integrate solutions, contribute to CI/CD, and support documentation and agile product development.

Data Analyst

May 2024 – Present

Center for Community Engagement, Environment Justice, and Health (CEEJH)

College Park, MD

- Managed real-time air quality data from IoT sensors (e.g., PurpleAir, C-12) and processed CSV and JSON files using Python and SQL for analysis.
- Contributed to the research, analysis and writing of a scientific paper that studied air pollution disparities using low-cost sensors in underserved communities, providing data to support environmental justice initiatives.

Research Assistant

Feb 2024 – May 2024

Computation Discovery and Design of New Material - Mo Research Group

College Park, MD

- Explored Airflow to schedule batch simulations and preprocess large-scale lithium compound datasets.
- Applied unsupervised learning techniques, including KMeans with the Elbow Method and Agglomerative Clustering, to analyze and cluster lithium compounds, leveraging machine learning models for DFT calculations to improve accuracy and efficiency in compound analysis, inspired by recent academic research.

PROJECTS

TerpMatch

- Designed a university dating app, securing \$1,500 in seed funding from University of Maryland.
- Developed key features such as real-time chat and account registration pages using Flutter. Successfully integrated Firebase for secure authentication and streamlined data storage by connecting the app to Firestore.

Terpiez

- Developed a game allowing players to capture virtual creatures on a campus map, track stats, and view real-time creature locations using GPS and GlobalProtect VPN integration.
- Implemented secure authentication with Redis and designed a seamless Flutter-based interface for enhanced gameplay and user experience.

MindPathways

- Developed a mental healthcare platform using Flask, React, and PostgreSQL with features like session scheduling and secure data storage, and utilized OpenAI's API to power a chatbot for empathetic, real-time responses.
- Used SQL queries and OLTP patterns for transactional operations, while designing JSON APIs to interact with client-side apps.
- Designed a scalable architecture with a RESTful API backend and a responsive React frontend, ensuring seamless interaction between psychology students, professionals, and users.